STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 12: Kinetic Theory

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

What is an ideal gas model? [1]
Ideal gas equation? [1]
Mean translational kinetic energy? [1]
Origin of gas pressure? [1]
Monatomic degrees of freedom? [1]
Effect of lowering pressure on mean free path? [1]
Molar C_v of monatomic gas? [1]
Rms speed relation? [1]

At fixed T, volume halves: pressure? [2]
27 deg C in Kelvin? [2]
Mean kinetic energy at 300K, k=1.38x10^-23? [2]
Temperature becomes 4T: rms speed factor? [2]
Internal energy of n mol monatomic gas? [2]
Two main ideal gas assumptions? [2]

Rms speed of M=.028,T=300K,R=8.31? [3]
1 mol monatomic gas warms 20K, C_v=3R/2: DeltaU? [3]
At fixed T, pressure triples: volume factor? [3]
State equipartition prediction for monatomic gas. [3]

Compare rms speeds for molar masses .004 and .016 kg/mol at same T. [4]
f=5, n=2 mol, T=300K: internal energy? [4]

End of paper - check working, units, signs and significant figures.